

[4] Retrieval-Augmented Generation for Multiple-Choice Questions and Answers Generation

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 - **Venue & Year:** Procedia Computer Science (2025), 259, 504–511
 - **Category:** LMS Question Generation
 - **Associated Full Paper PDF:** Reference [4] (Procedia Computer Science)
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1-Minute Executive Takeaway

Key Finding: Validates PDF-to-quiz generation in an LMS, establishing the baseline workflow that Kwiz optimizes.

Core Research Objective

To investigate automated multiple-choice question generation from PDF course resources and integrate question delivery within the Ample Learning Management System.

Methodology & Architecture

Extracted text from lecture PDFs, chunked content into passages, embedded text into a vector database, and prompted an LLM to generate questions and corresponding options.

Key Empirical Findings & Evidence

Confirmed that grounding generation on PDF textbooks significantly improves topical alignment, but full re-indexing of course documents upon each upload introduces noticeable server delays.

Direct Relevance to Our Kwiz Paper

Cited in Section 1 and Section 2.2 to demonstrate that LMS-integrated RAG question generation has precedent. We advance beyond Pradeesh et al. by introducing incremental content-hash caching, programming code execution validation, and native Moodle Question Bank persistence.